

SAMUEL GREENE

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Education

University of California, Davis

Expected June 2027

Bachelor of Science, Computer Engineering & Science — GPA: 3.46

Davis, CA

- **Relevant Coursework:** Circuits I-II, PHY 9A-D (Classical, Electromagnetic, and Modern Nuclear/Quantum Physics), Embedded Systems, Operating Systems, Advanced Computer Architecture, Theory of Computation, Algorithms & Data Structures, Machine Programming

Research Experience

CoreView Project — UC Davis / University of Washington

June 2026 – Present

Systems Engineer

Davis, CA / Seattle, WA

- Develop automated software test utilities and concurrent data-ingestion systems using **Python** and **C++** to process high-volume sensor and imaging data.
- Own the project's **AWS and RunPod infrastructure**, navigating **Linux command-line environments** to deploy GPU-based pipelines using EC2, S3, and ECR.
- Provision edge-computing and GPU compute resources, deploying containerized inference workloads with **Docker**, PyTorch, TensorFlow, and CUDA-accelerated OpenCV.
- Reduced medical microscopy image-processing time from 2–3 days to 15–30 minutes by moving computational workloads to GPU-based processing pipelines.

UCD Medical Center — Levenson Laboratory

Sep 2024 – June 2026

Research Programmer

Davis, CA

- Built **FIBI Post Processing**, an in-lab desktop application that accelerated microscopy image post-processing pipelines by more than 2×, improving researcher throughput.
- Lead developer on **Maggie**, the primary software interface to custom microscopy hardware; owned the interactive **ROI-selection** system from UI design through backend integration.
- Integrated **OpenCV** feature detection and automated image-analysis tools, including color correction, multi-image overlays, and feature detection to reduce manual review.

Projects

GPS Backpacking Telemetry Display

C++ | UART/I2C | Hardware Verification | OLED

- Engineered a real-time NMEA telemetry display for backpacking, utilizing a non-blocking *millis()* execution loop in C++ for 4Hz UI redraws and continuous GPS data parsing.
- Implemented dynamic pagination for navigation and telemetry data, alongside real-time signal integrity (HDOP) and satellite lock monitoring.
- Troubleshooted and verified hardware performance using **Saleae logic analyzers** and **physical oscilloscopes** to validate UART/I2C bus timing, signal integrity, and stable data delivery.

Bicycle Speed & GPS Tracking HUD

Embedded C | SPI/I2C | Sensor Fusion

- Designed and built a real-time embedded heads-up display streaming speed and trip statistics via GPS/IMU sensor fusion.
- Wrote bare-metal firmware for concurrent peripheral drivers and interrupt-driven data capture under strict power and memory constraints.

Programming Language Tokenizer & Lexer

C++ | Linux Toolchain | Makefile

- Designed a tokenizer and lexer for a custom programming language grounded in formal **Theory of Computation** principles, using Pratt parsing techniques for extensibility.

Technical Skills

Hardware & Testing: Physical oscilloscopes, Saleae logic analyzers, schematics reading, electronic circuit theory

Embedded & Systems: C/C++, Embedded C, SPI, I2C, UART, GPIO, bare-metal MCU programming, sensor fusion

Software & Scripting: Python, Linux/Unix command-line, Git, Bash, C#, Rust, JavaScript, Java

Edge & Compute: Docker, CUDA, OpenCV, PyTorch, TensorFlow, Azure, AWS (EC2, S3), RunPod